

Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.53)

Metric

$$ds^2 = -dt \otimes dt + \left(t^{\frac{4}{3}} \cos(\psi) + \frac{2}{3} \right) dx \otimes dx + \left(t^{\frac{4}{3}} \cos\left(\frac{2}{3}\pi\right) \cos(\psi) - \frac{4}{3} \sin\left(\frac{2}{3}\pi\right) \sin(\psi) + \frac{2}{3} \right) dy \otimes dy + \left(t^{\frac{4}{3}} \cos\left(\frac{4}{3}\pi\right) \cos(\psi) - \frac{4}{3} \sin\left(\frac{4}{3}\pi\right) \sin(\psi) + \frac{2}{3} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$p_{alpha} = \frac{2}{3} \cos\left(\frac{2}{3}\alpha\pi - \frac{2}{3}\pi + \psi\right) + \frac{1}{3}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{1}{2}\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos(\psi)\right) dx$$

$$n = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{1}{2}\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos(\psi)\right) dx$$

$$m = \left(\frac{1}{2}\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos\left(\frac{2}{3}\pi + \psi\right)\right) dy + \left(\frac{1}{2}i\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos\left(\frac{4}{3}\pi + \psi\right)\right) dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos\left(\frac{2}{3}\pi + \psi\right)\right) dy + \left(-\frac{1}{2}i\sqrt{2}t^{\frac{1}{3}}t^{\frac{2}{3}}\cos\left(\frac{4}{3}\pi + \psi\right)\right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2}}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos(\psi)} \right) dx$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\sqrt{2}}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos(\psi)} \right) dx$$

$$m = \left(\frac{\sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{2}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{2}{3} \pi\right) \cos(\psi)} \right) dy + \left(\frac{i \sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{4}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{4}{3} \pi\right) \cos(\psi)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{2}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{2}{3} \pi\right) \cos(\psi)} \right) dy + \left(-\frac{i \sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{4}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{4}{3} \pi\right) \cos(\psi)} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2}}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos(\psi)} \partial_x$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2}}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos(\psi)} \partial_x$$

$$\delta = \frac{\sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{2}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{2}{3} \pi\right) \cos(\psi)} \partial_y + \frac{i \sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{4}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{4}{3} \pi\right) \cos(\psi)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{2}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{2}{3} \pi\right) \cos(\psi)} \partial_y - \frac{i \sqrt{2} t^{\frac{2}{3}} \sin\left(\frac{4}{3} \pi\right) \sin(\psi)}{2 t^{\frac{1}{3}} t^{\frac{2}{3}} \cos\left(\frac{4}{3} \pi\right) \cos(\psi)} \partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{6t} + \frac{\sqrt{2} \cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{6t} - \frac{\sqrt{2} \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{6t} - \frac{\sqrt{2} \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{6t} + \frac{\sqrt{2}}{6t}$$

$$\sigma = -\frac{\sqrt{2} \cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{6t} + \frac{\sqrt{2} \cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{6t} + \frac{\sqrt{2} \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{6t} - \frac{\sqrt{2} \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{6t}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2} \cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{6t} - \frac{\sqrt{2} \cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{6t} + \frac{\sqrt{2} \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{6t} + \frac{\sqrt{2} \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{6t} - \frac{\sqrt{2}}{6t}$$

$$\lambda = \frac{\sqrt{2} \cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{6t} - \frac{\sqrt{2} \cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{6t} - \frac{\sqrt{2} \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{6t} + \frac{\sqrt{2} \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{6t}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2} \cos(\psi)}{6t} - \frac{\sqrt{2}}{12t}$$

$$\gamma = \frac{\sqrt{2} \cos(\psi)}{6t} + \frac{\sqrt{2}}{12t}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & -\frac{2 \cos\left(\frac{2}{3}\pi\right)^2 \cos(\psi)^2}{9t^2} + \frac{2 \cos(\psi)^2 \sin\left(\frac{4}{3}\pi\right)^2}{9t^2} + \frac{2 \cos\left(\frac{4}{3}\pi\right) \cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} + \frac{2 \cos\left(\frac{2}{3}\pi\right) \cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} + \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)^2}{9t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)^2}{9t^2} \\ & - \frac{\cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} + \frac{\cos\left(\frac{2}{3}\pi\right)^2}{9t^2} + \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{9t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{9t^2} - \frac{\sin\left(\frac{4}{3}\pi\right)^2}{9t^2} - \frac{\sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} + \frac{\cos(\psi)}{9t^2} + \frac{1}{18t^2}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned}\Phi_{02} = & \frac{2 \cos(\psi)^2 \sin\left(\frac{4}{3}\pi\right)^2}{9t^2} - \frac{2 \cos(\psi)^2 \sin\left(\frac{2}{3}\pi\right)^2}{9t^2} + \frac{2 \cos\left(\frac{4}{3}\pi\right) \cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{2 \cos\left(\frac{2}{3}\pi\right) \cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} \\ & - \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)^2}{9t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)^2}{9t^2} + \frac{\cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\sin\left(\frac{4}{3}\pi\right)^2}{9t^2} + \frac{\sin\left(\frac{2}{3}\pi\right)^2}{9t^2}\end{aligned}$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & -\frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\cos\left(\frac{4}{3}\pi\right) \cos\left(\frac{2}{3}\pi\right) \sin(\psi)^2}{9t^2} + \frac{\sin\left(\frac{4}{3}\pi\right) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)^2}{9t^2} \\ & + \frac{\cos\left(\frac{4}{3}\pi\right) \cos\left(\frac{2}{3}\pi\right)}{9t^2} + \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{18t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{18t^2} - \frac{\sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{18t^2} - \frac{\sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{18t^2} + \frac{\sin(\psi)^2}{9t^2} + \frac{\cos(\psi)}{18t^2} - \frac{1}{36t^2}\end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}\Phi_{20} = & \frac{2 \cos(\psi)^2 \sin\left(\frac{4}{3}\pi\right)^2}{9t^2} - \frac{2 \cos(\psi)^2 \sin\left(\frac{2}{3}\pi\right)^2}{9t^2} + \frac{2 \cos\left(\frac{4}{3}\pi\right) \cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{2 \cos\left(\frac{2}{3}\pi\right) \cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} \\ & - \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)^2}{9t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)^2}{9t^2} + \frac{\cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9t^2} - \frac{\sin\left(\frac{4}{3}\pi\right)^2}{9t^2} + \frac{\sin\left(\frac{2}{3}\pi\right)^2}{9t^2}\end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned} \Phi_{22} = & -\frac{2 \cos(\frac{2}{3}\pi)^2 \cos(\psi)^2}{9t^2} + \frac{2 \cos(\psi)^2 \sin(\frac{4}{3}\pi)^2}{9t^2} + \frac{2 \cos(\frac{4}{3}\pi) \cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} + \frac{2 \cos(\frac{2}{3}\pi) \cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} + \frac{\cos(\frac{4}{3}\pi) \cos(\psi)^2}{9t^2} + \frac{\cos(\frac{2}{3}\pi) \cos(\psi)^2}{9t^2} \\ & - \frac{\cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} - \frac{\cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} + \frac{\cos(\frac{2}{3}\pi)^2}{9t^2} + \frac{\cos(\frac{4}{3}\pi) \cos(\psi)}{9t^2} + \frac{\cos(\frac{2}{3}\pi) \cos(\psi)}{9t^2} - \frac{\sin(\frac{4}{3}\pi)^2}{9t^2} - \frac{\sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} - \frac{\sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} + \frac{\cos(\psi)}{9t^2} + \frac{1}{18t^2} \end{aligned}$$

$$\Lambda = \frac{\cos(\frac{4}{3}\pi)\cos(\frac{2}{3}\pi)\cos(\psi)^2}{27t^2} - \frac{2\cos(\psi)^2\sin(\frac{4}{3}\pi)^2}{27t^2} - \frac{\cos(\psi)^2\sin(\frac{4}{3}\pi)\sin(\frac{2}{3}\pi)}{27t^2} - \frac{2\cos(\psi)^2\sin(\frac{2}{3}\pi)^2}{27t^2} - \frac{2\cos(\frac{4}{3}\pi)\cos(\psi)\sin(\frac{4}{3}\pi)\sin(\psi)}{27t^2} \\ - \frac{\cos(\frac{2}{3}\pi)\cos(\psi)\sin(\frac{4}{3}\pi)\sin(\psi)}{27t^2} - \frac{\cos(\frac{4}{3}\pi)\cos(\psi)\sin(\frac{2}{3}\pi)\sin(\psi)}{27t^2} - \frac{2\cos(\frac{2}{3}\pi)\cos(\psi)\sin(\frac{2}{3}\pi)\sin(\psi)}{27t^2} + \frac{\cos(\frac{4}{3}\pi)\cos(\psi)^2}{27t^2} \\ + \frac{\cos(\frac{2}{3}\pi)\cos(\psi)^2}{27t^2} - \frac{\cos(\psi)\sin(\frac{4}{3}\pi)\sin(\psi)}{27t^2} - \frac{\cos(\psi)\sin(\frac{2}{3}\pi)\sin(\psi)}{27t^2} + \frac{\cos(\frac{4}{3}\pi)\cos(\psi)}{54t^2} + \frac{\cos(\frac{2}{3}\pi)\cos(\psi)}{54t^2} \\ + \frac{\cos(\psi)^2}{9t^2} + \frac{\sin(\frac{4}{3}\pi)^2}{27t^2} + \frac{\sin(\frac{4}{3}\pi)\sin(\frac{2}{3}\pi)}{27t^2} + \frac{\sin(\frac{2}{3}\pi)^2}{27t^2} - \frac{\sin(\frac{4}{3}\pi)\sin(\psi)}{54t^2} - \frac{\sin(\frac{2}{3}\pi)\sin(\psi)}{54t^2} + \frac{\cos(\psi)}{54t^2} - \frac{1}{36t^2}$$

Weyl Tensor Components

$$\begin{aligned} \Psi_0 = & -\frac{2 \cos(\psi)^2 \sin(\frac{4}{3}\pi)^2}{9t^2} + \frac{2 \cos(\psi)^2 \sin(\frac{2}{3}\pi)^2}{9t^2} - \frac{2 \cos(\frac{4}{3}\pi) \cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} + \frac{2 \cos(\frac{2}{3}\pi) \cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} - \frac{\cos(\frac{4}{3}\pi) \cos(\psi)^2}{9t^2} + \frac{\cos(\frac{2}{3}\pi) \cos(\psi)^2}{9t^2} \\ & + \frac{\cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} - \frac{\cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} - \frac{\cos(\frac{4}{3}\pi) \cos(\psi)}{9t^2} + \frac{\cos(\frac{2}{3}\pi) \cos(\psi)}{9t^2} + \frac{\sin(\frac{4}{3}\pi)^2}{9t^2} - \frac{\sin(\frac{2}{3}\pi)^2}{9t^2} + \frac{\sin(\frac{4}{3}\pi) \sin(\psi)}{9t^2} - \frac{\sin(\frac{2}{3}\pi) \sin(\psi)}{9t^2} \end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & -\frac{2 \cos(\frac{4}{3}\pi) \cos(\frac{2}{3}\pi) \cos(\psi)^2}{27 t^2} - \frac{2 \cos(\psi)^2 \sin(\frac{4}{3}\pi)^2}{27 t^2} + \frac{2 \cos(\psi)^2 \sin(\frac{4}{3}\pi) \sin(\frac{2}{3}\pi)}{27 t^2} - \frac{2 \cos(\psi)^2 \sin(\frac{2}{3}\pi)^2}{27 t^2} \\ & - \frac{2 \cos(\frac{4}{3}\pi) \cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{27 t^2} + \frac{2 \cos(\frac{2}{3}\pi) \cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{27 t^2} + \frac{2 \cos(\frac{4}{3}\pi) \cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{27 t^2} \\ & - \frac{2 \cos(\frac{2}{3}\pi) \cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{27 t^2} + \frac{\cos(\frac{4}{3}\pi) \cos(\psi)^2}{27 t^2} + \frac{\cos(\frac{2}{3}\pi) \cos(\psi)^2}{27 t^2} - \frac{\cos(\psi) \sin(\frac{4}{3}\pi) \sin(\psi)}{27 t^2} - \frac{\cos(\psi) \sin(\frac{2}{3}\pi) \sin(\psi)}{27 t^2} \\ & - \frac{\cos(\frac{4}{3}\pi) \cos(\psi)}{27 t^2} - \frac{\cos(\frac{2}{3}\pi) \cos(\psi)}{27 t^2} + \frac{\sin(\frac{4}{3}\pi)^2}{27 t^2} - \frac{2 \sin(\frac{4}{3}\pi) \sin(\frac{2}{3}\pi)}{27 t^2} + \frac{\sin(\frac{2}{3}\pi)^2}{27 t^2} + \frac{\sin(\frac{4}{3}\pi) \sin(\psi)}{27 t^2} + \frac{\sin(\frac{2}{3}\pi) \sin(\psi)}{27 t^2} + \frac{2 \cos(\psi)}{27 t^2} \end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}\Psi_4 = & -\frac{2 \cos(\psi)^2 \sin\left(\frac{4}{3}\pi\right)^2}{9 t^2} + \frac{2 \cos(\psi)^2 \sin\left(\frac{2}{3}\pi\right)^2}{9 t^2} - \frac{2 \cos\left(\frac{4}{3}\pi\right) \cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9 t^2} + \frac{2 \cos\left(\frac{2}{3}\pi\right) \cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9 t^2} - \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)^2}{9 t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)^2}{9 t^2} \\ & + \frac{\cos(\psi) \sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9 t^2} - \frac{\cos(\psi) \sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9 t^2} - \frac{\cos\left(\frac{4}{3}\pi\right) \cos(\psi)}{9 t^2} + \frac{\cos\left(\frac{2}{3}\pi\right) \cos(\psi)}{9 t^2} + \frac{\sin\left(\frac{4}{3}\pi\right)^2}{9 t^2} - \frac{\sin\left(\frac{2}{3}\pi\right)^2}{9 t^2} + \frac{\sin\left(\frac{4}{3}\pi\right) \sin(\psi)}{9 t^2} - \frac{\sin\left(\frac{2}{3}\pi\right) \sin(\psi)}{9 t^2}\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"